

LIOUA ZEDDAM

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Full Stack Developer · Django · React · Flutter · PostgreSQL

Summary

Full Stack Developer who has built production-grade web **and cross-platform mobile** applications using **Django REST Framework, React, TypeScript**, and **Flutter** on **PostgreSQL**, with additional backend experience in **Node.js** from independent project work. Specializes in high-performance backend architecture — REST API design, real-time WebSocket systems, Redis caching, transactional integrity, and containerized VPS deployments. Experienced owning the full stack end-to-end and working directly with clients and cross-functional stakeholders in both freelance and institutional environments.

Work Experience

Backend Architect & Admin Frontend Developer · Freelance

Dec 2025 – Present

Rezervitoo (Client Project) | Backend + Admin Panel (sole owner), Mobile + Website (collaborator)

Algiers, Algeria

- Architected a multi-provider enterprise travel and booking platform supporting **4 provider types** (Hotels, Hostels, Agencies, Hosts) using **Django and DRF**, deployed via **Docker Compose, Nginx, and Cloudflare SSL Full (Strict)**.
- Designed, implemented, and Swagger/OpenAPI documented **100+ REST API endpoints** across **7 resource domains** (auth, users, listings, bookings, reviews, reports, notifications) with **RBAC enforced across all routes**.
- Reduced API response time from **176ms to 38ms** on repeat requests by implementing **Redis caching** with TanStack Query client-side synchronization, delivering near-instant data fetching on second load.
- Engineered a real-time bi-directional notification system using **Django Channels (ASGI), WebRTC**, and **Firestore Cloud Messaging (FCM)**, covering web push and mobile delivery across all user role types — secured via custom **JWT WebSocket middleware**.
- Implemented a **bidirectional review and report system** — users and providers mutually rate, comment, and submit image evidence — with moderation hooks feeding into admin panel deep-linked management views.
- Implemented **Polymorphic Model** architecture to unify **4 accommodation types** under a single schema, eliminating schema duplication and reducing provider onboarding surface by **60%**.
- Engineered full **RTL/LTR UI mirroring** for Arabic, English, and French using **django-modeltranslation** and CSS logical properties, delivering a layout that structurally adapts to language direction.
- Built a React Typescript admin panel with **millisecond-range load times** and deep-linked management views, with **Google OAuth2** ID token verification, **Brevo** transactional email, and progressive provider onboarding flows.

Full Stack Developer · Apprenticeship

Jan 2024 – Apr 2025

Faderco · Dual-Study Program

Setif, Algeria

- Delivered an internal **Recruitment Portal** using Django and React, digitizing the full HR pipeline from candidate intake through offer — eliminating manual spreadsheet tracking for **50+ candidates per cycle**.
- Integrated **WebRTC and StreamAPI** to ship the company's first remote interview capability, eliminating up to **4 hours of round-trip travel** per interview for candidates commuting to the company's isolated Setif location.
- Built a **live notification system using WebRTC** to push real-time updates to HR staff and candidates throughout the interview and evaluation pipeline, replacing manual follow-up with instant in-app alerts.
- Engineered a **worker performance evaluation module**, replacing paper-based assessment workflows and saving HR an estimated **2+ hours of manual input** per evaluation cycle.
- Implemented **Role-Based Access Control (RBAC)** using Django permissions and **Google OAuth2** to enforce data privacy over sensitive records across **3+ distinct roles**.
- Managed the full project lifecycle solo across **10 months** — from requirements gathering with HR stakeholders through production deployment — bridging HR and District Managers with zero prior internal tooling.

Projects

BrewPhoria — Coffee Ordering Platform (Mobile + API) · *Source Code***Flutter** · **Node.js** · **TypeScript** · **PostgreSQL**

- Built a cross-platform **Flutter** client (**Riverpod**, feature-first architecture) backed by a layered **Node.js** / **Express** / **TypeScript** REST API with **Prisma** and **PostgreSQL** — **40+ endpoints** across **11 resource domains**.
- Engineered an **atomic checkout** that creates the order, decrements stock, clears the cart, and moves loyalty points inside a single **Prisma transaction** — with all pricing, discounts, and points recomputed server-side to keep the client untrusted.
- Designed an **offline-first guest cart** persisted in **Hive** that merges conflict-free into the account on sign-in (surviving cold starts via a pending-merge flag), removing the pre-signup friction that blocks conversion.
- Implemented **Redis caching** with per-query cache keys and pattern-based invalidation on writes, a server-side product **modifier/pricing engine**, and a tiered **loyalty programme** (earn multipliers, 100 pts = \$1 redemption) backed by an append-only ledger.
- Integrated a **tool-augmented AI barista (Google Gemini)** that resolves product tags to real catalogue items, plus **Firestore** Auth + **FCM** push; containerized with **Docker Compose** and documented with architecture diagrams and a full API reference.

FitGuild — E-Commerce Platform · *Live Site | Source Code* **Django** · **React** · **PostgreSQL**

- Achieved **sub-2s load times** across all product catalog pages via lazy loading, image optimization, and multi-layer Nginx caching — confirmed through Core Web Vitals profiling.
- Integrated **Stripe Checkout with webhooks** handling the full post-purchase event loop server-side: secure payments, automated order status transitions, and real-time inventory decrement.
- Shipped a complete commerce feature set end-to-end: **product variants, SKU management, cart, wishlist, order history, and review system** — REST API design through React UI with zero third-party commerce dependencies.
- Delivered real-time order status push via **Django Channels ASGI consumers** over persistent WebSocket connections, replacing polling and eliminating unnecessary server load.

Cognitive Training Platform · *Source Code* **Django** · **React** · **PostgreSQL**

- Engineered a gamified cognitive platform featuring **7 categories** and **14 interactive browser-based games**, with dynamic difficulty scaling, XP progression, and a multi-dimensional ranking system.
- Designed a real-time multiplayer infrastructure supporting concurrent users — including **global chat, private messaging, and live leaderboards** — using **Django Channels WebSockets** and **WebRTC** communication.
- Implemented a **performance scoring system** combining accuracy, streaks, and response time to generate detailed user analytics and progression insights across categories and games.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, Dart, SQL
Backend: Django, Django REST Framework, Node.js, Express, Prisma, PostgreSQL, Redis, Celery
Mobile: Flutter, Riverpod, Freezed, Hive, go_router, Dio
Frontend: React.js, Tailwind CSS, TanStack Query, Redux
Infrastructure / DevOps: Docker, Docker Compose, Nginx, Linux/VPS, Git, Github Actions
Real-Time & AI: Django Channels (ASGI), WebSockets, WebRTC, Firebase (Auth/FCM), Google Gemini
Cloud & APIs: Google Cloud (IAM, OAuth2), Stripe, Cloudflare, Brevo, Swagger/OpenAPI
Architecture: REST API Design, RBAC, Layered / Service Architecture, Polymorphic Models, Offline-First, i18n/RTL
Testing: PyTest, Jest, Supertest

Education

Diploma in Computer Science — Database Foundations **Oct 2022 – Apr 2025**
Baaich Abdelkader Institute · Ranked 1st in Cohort *Setif, Algeria*

- Final GPA: **14.79/20** | Dual-Study Program with 2.5-year integrated placement at Faderco

Advanced React & Frontend Engineering Certification **2025**
Codecademy · Remote

Languages

Arabic: Native **English:** C2 / Professional **French:** B2 / Intermediate